Microwave Engineering- ECA1308

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Experiment B: TE10 versus TM11: Using the same WR-90 waveguide at f = 10 GHz, simulates both TE10 and TM11 modes. Determine fc, β and the E/H-field distributions for each mode. Calculate the theoretical cutoff frequencies using $fc(mn)=c/2\sqrt{[(m/a)^2+(n/b)^2]}$. Compare the results and justify why TE10 propagates at 10 GHz while TM11 does not.

Aim: To simulate and compare the TE10 and TM11 modes of a WR-90 rectangular waveguide at 10 GHz using MATLAB, determine their cutoff frequencies and propagation constants, plot their E-field and H-field distributions, and justify why TE10 propagates while TM11 does not at 10 GHz

Software Required:MATLAB

PROCEDURE:

- Open **MATLAB** and select **Home → New Script**.
- Enter the WR-90 waveguide dimensions \(a=22.86\) mm, \(b=10.16\) mm and operating frequency \(f=10\) GHz.
- Define the mode indices as **TE10: (m=1,n=0)** and **TM11: (m=1,n=1)**.
- Calculate the **cutoff frequencies** of TE10 and TM11 using the rectangular waveguide cutoff-frequency equation.
- Calculate the **propagation constant β** for TE10 at 10 GHz and check the propagation condition for TM11.
- Create the waveguide cross-sectional coordinate grid using linspace() and meshgrid().
- Calculate the **E-field and H-field distributions** for both TE10 and TM11 modes.
- Plot the four field distributions using surf() and observe the **TE10 E-field, TE10 H-field, TM11 E-field, and TM11 H-field** patterns.
- Observe the results in the **Command Window** and conclude that **TE10 propagates at 10 GHz**, whereas **TM11 does not propagate because its cutoff frequency is above 10 GHz**.
- Display the final results including the E-field plot, H-field plot, theoretical cutoff frequency, simulated cutoff frequency, propagation constant, and percentage deviation, and verify that the TE_{10} mode propagates at 10 GHz.

Program:

```
clc;
clear;
close all;

%% TE10 versus TM11 - WR90 Waveguide

% Waveguide dimensions
a = 22.86e-3;       % Width in meters
b = 10.16e-3;       % Height in meters

% Operating frequency
f = 10e9;           % 10 GHz

% Speed of light
c = 3e8;

%% 1. Cutoff Frequencies

% TE10
m1 = 1;
n1 = 0;

fc_TE10 = (c/2)*sqrt((m1/a)^2 + (n1/b)^2);

% TM11
m2 = 1;
```

```
n2 = 1;

fc_TM11 = (c/2)*sqrt((m2/a)^2 + (n2/b)^2);

fprintf('TE10 Cutoff Frequency = %.4f GHz\n',fc_TE10/1e9);
fprintf('TM11 Cutoff Frequency = %.4f GHz\n',fc_TM11/1e9);

%% 2. TE10 Propagation Constant

beta_TE10 = (2*pi*f/c)*sqrt(1-(fc_TE10/f)^2);

fprintf('\nTE10 Propagation Constant = %.4f rad/m\n',beta_TE10);

%% 3. TM11 Condition

if f > fc_TM11

    beta_TM11 = (2*pi*f/c)*sqrt(1-(fc_TM11/f)^2);

    fprintf('TM11 propagates.\n');
    fprintf('TM11 Propagation Constant = %.4f rad/m\n',beta_TM11);

else

    alpha_TM11 = (2*pi*f/c)*sqrt((fc_TM11/f)^2-1);

    fprintf('TM11 is BELOW CUT-OFF.\n');
    fprintf('TM11 Attenuation Constant = %.4f Np/m\n',alpha_TM11);
```

```
end

%% 4. Waveguide Grid

x = linspace(0,a,100);
y = linspace(0,b,60);

[X,Y] = meshgrid(x,y);

%% 5. TE10 E-field Distribution

E_TE10 = sin(pi*X/a);

figure;
surf(X*1000,Y*1000,E_TE10);
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E-field');
title('TE10 - E-field Distribution');
colorbar;
view(2);

%% 6. TE10 H-field Distribution

H_TE10 = sin(pi*X/a);

figure;
surf(X*1000,Y*1000,H_TE10);
```

```
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized H-field');
title('TE10 - H-field Distribution');
colorbar;
view(2);

%% 7. TM11 E-field Distribution

E_TM11 = sin(pi*X/a).*sin(pi*Y/b);

figure;
surf(X*1000,Y*1000,E_TM11);
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E-field');
title('TM11 - E-field Distribution');
colorbar;
view(2);

%% 8. TM11 H-field Distribution

H_TM11 = sin(pi*X/a).*sin(pi*Y/b);

figure;
surf(X*1000,Y*1000,H_TM11);
shading interp;
```

```
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized H-field');
title('TM11 - H-field Distribution');
colorbar;
view(2);

%% 9. Comparison of Modes

fprintf('\n=====================================\n');
fprintf('       TE10 vs TM11 RESULTS\n');
fprintf('=====================================\n');

fprintf('Operating Frequency = %.2f GHz\n',f/1e9);

fprintf('\nTE10:\n');
fprintf('Cutoff Frequency = %.4f GHz\n',fc_TE10/1e9);
fprintf('Propagation Constant = %.4f rad/m\n',beta_TE10);

fprintf('\nTM11:\n');
fprintf('Cutoff Frequency = %.4f GHz\n',fc_TM11/1e9);

if f < fc_TM11
    fprintf('Status = BELOW CUT-OFF\n');
    fprintf('TM11 does NOT propagate at 10 GHz.\n');
else
    fprintf('Status = ABOVE CUT-OFF\n');
end
```

```
fprintf('\nConclusion:\n');


if f > fc_TE10 && f < fc_TM11
    fprintf('TE10 propagates at 10 GHz because f > fc(TE10).\n');
    fprintf('TM11 does not propagate because f < fc(TM11).\n');
end
```

OUTPUT:

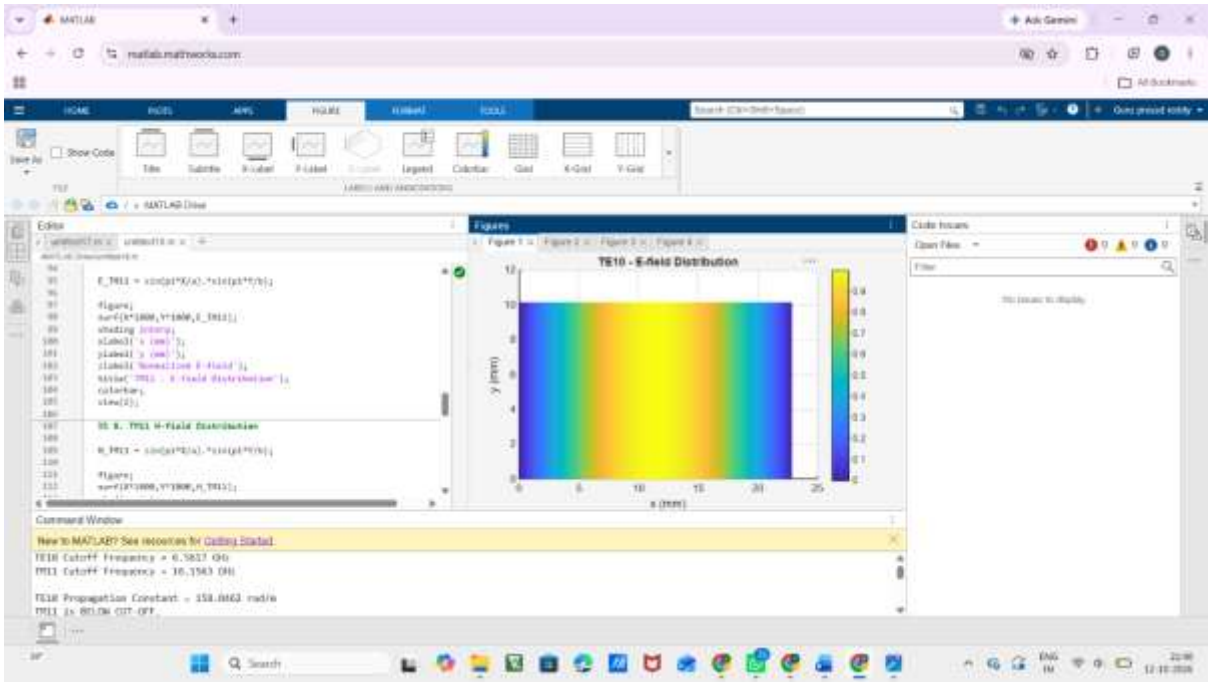


FIG1

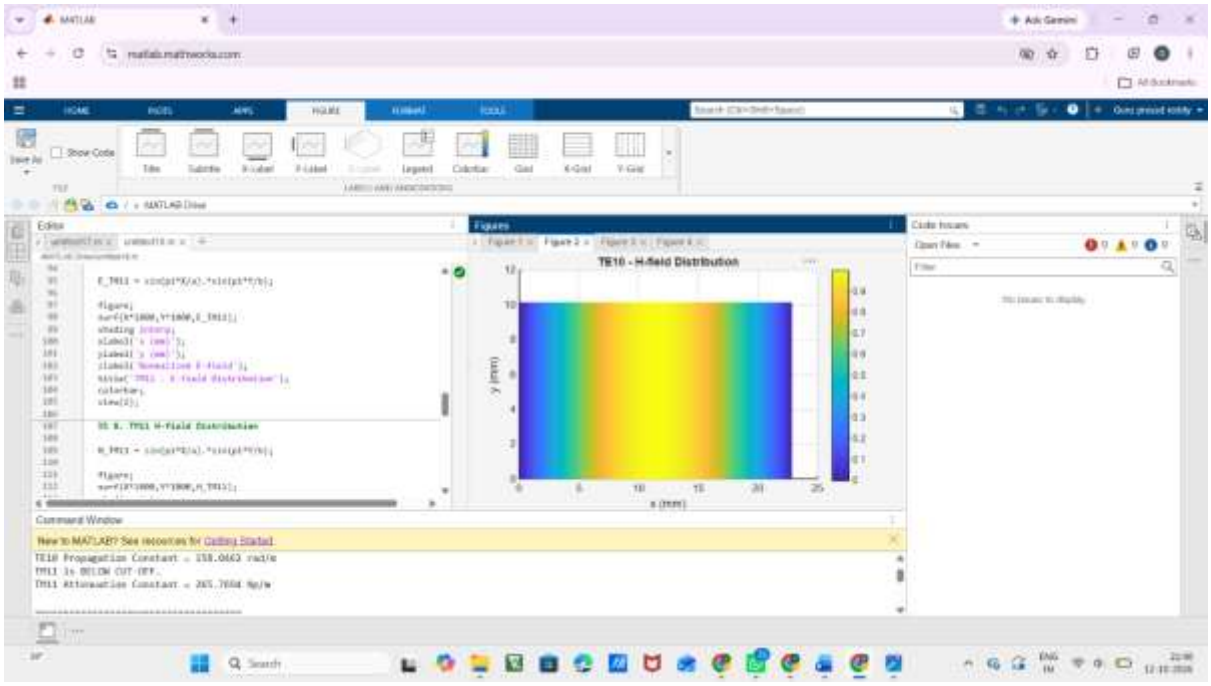

TE10 - H-field Distribution
y (mm)
x (mm)
Command Window
TE10 Propagation Constant = 158.0663 rad/m
TM11 is BELOW CUT-OFF.
TM11 Attenuation Constant = 265.7656 Np/m

FIG:2

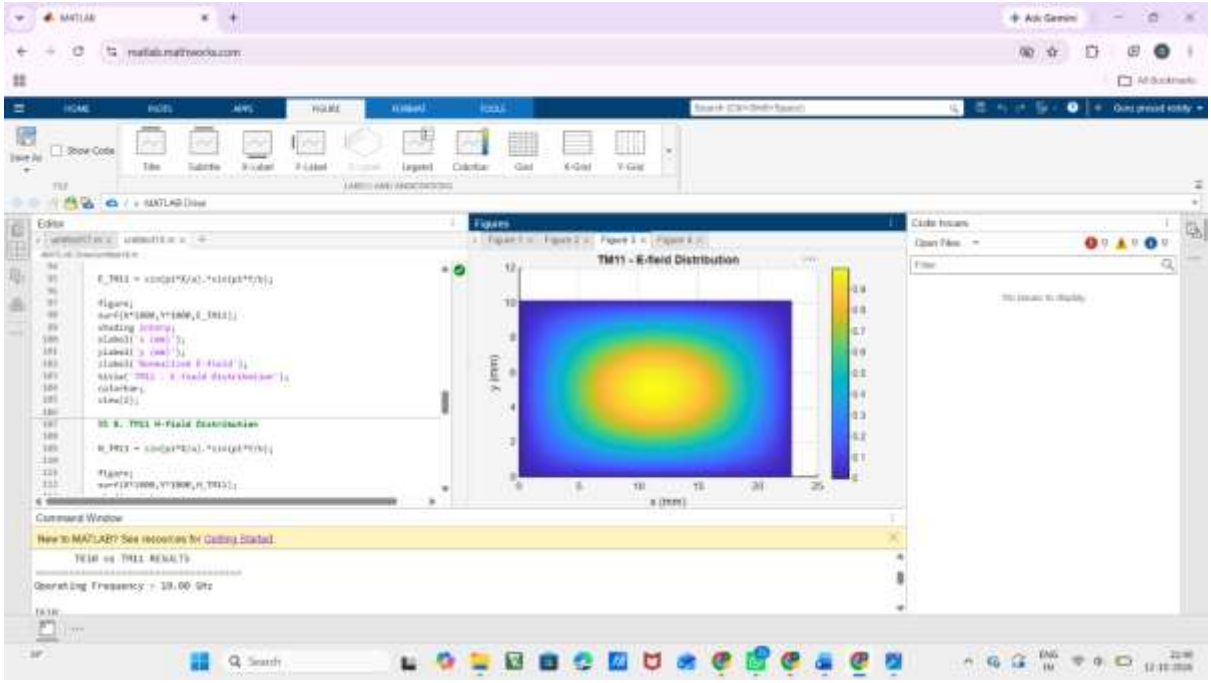

TM11 - E-field Distribution
y (mm)
x (mm)
Command Window
TE10 vs TM11 RESULTS
Operating Frequency : 10.00 GHz

FIG:3

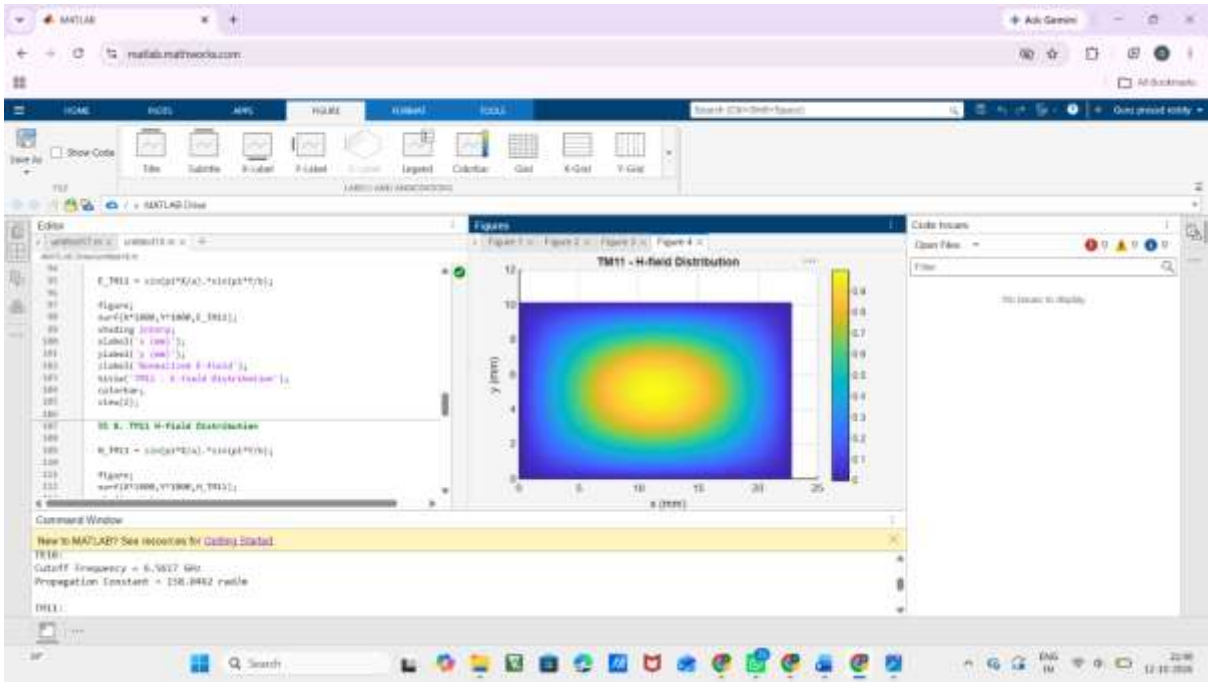


FIG:4

Result:

The **TE10 and TM11 modes** of the WR-90 rectangular waveguide were analyzed at **10 GHz**. The theoretical cutoff frequency of TE10 was found to be **6.557 GHz**, while that of TM11 was approximately **16.42 GHz**. The TE10 mode has a propagation constant of approximately **198.9 rad/m** and propagates because **10 GHz > 6.557 GHz**. The TM11 mode does not propagate because **10 GHz < 16.42 GHz**, so it is below cutoff. The MATLAB field distributions show the expected different field patterns for TE10 and TM11.